

GPS Locator communication protocol

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1. Communication protocols

introductory

This document defines the description of the application layer interface protocol for the vehicle GPS locator, positioning service platform. This interface protocol only applies to the interactive communication between the server platform and the positioning terminal.

2. Terminology, definitions

Terminology, abbreviations	English Meaning	Chinese Meaning
CMPP	China Mobile Peer to Peer	China Mobile P2P protocol

GPS	Global Positioning System	Global Positioning System (GPS)
GSM	Global System for Mobile Communication	Global System for Mobile Communications (GSM)
GPRS	General Packet Radio Service	Universal Wireless Packet Service (UWPS) (computing)
TCP	Transport Control Protocol	transmission control protocol
LBS	Location Based Services	assisted location service
IMEI	International Mobile Equipment Identity	International Mobile Equipment Identity (IMEI) (computing)
MCC	Mobile Country Code	Mobile subscriber country code
MNC	Mobile Network Code	Mobile network number
LAC	Location Area Code	location area code
Cell ID	Cell Tower ID	cell phone base station
UDP	User Datagram Protocol	User Datagram Protocol
SOS	Save Our Ship/Save Our Souls	distress signal
CRC	Cyclic Redundancy Check	cyclic redundancy check
NITZ	Network Identity and Time Zone.	time zones
GIS	Geographic Information System	Geographic Information System (GIS)

3. Ground rules

1. The GPRS connection is successfully established and the first login packet is sent to the server. If a response packet is received from the server within 5 seconds, the connection is considered to be normal and the positioning information (GPS, LBS packet) will be sent, and a status packet will be sent after 3 minutes to confirm the normal communication at regular intervals;

2. When the GPRS connection is not successfully established, the terminal cannot send the login information packet. When the GPRS connection fails 3 times, the terminal starts the timed restart function for 20 minutes. If the terminal successfully establishes a connection with the server and receives a packet from the server in response to the login information packet sent by the terminal within 20 minutes, the timed restart function is turned off and the terminal is not restarted, otherwise the terminal is restarted automatically after 20 minutes;

3. The server needs to return the response packet to the terminal after it receives the login information sent by the terminal. If the terminal does not receive the return packet from the server for more than 5 seconds after it sends the login information packet or the status information packet, it considers that the current connection is established abnormally, starts the function of GPS positioning data retransmission, disconnects the current GPRS connection, establishes a new GPRS connection and sends the login information packet;

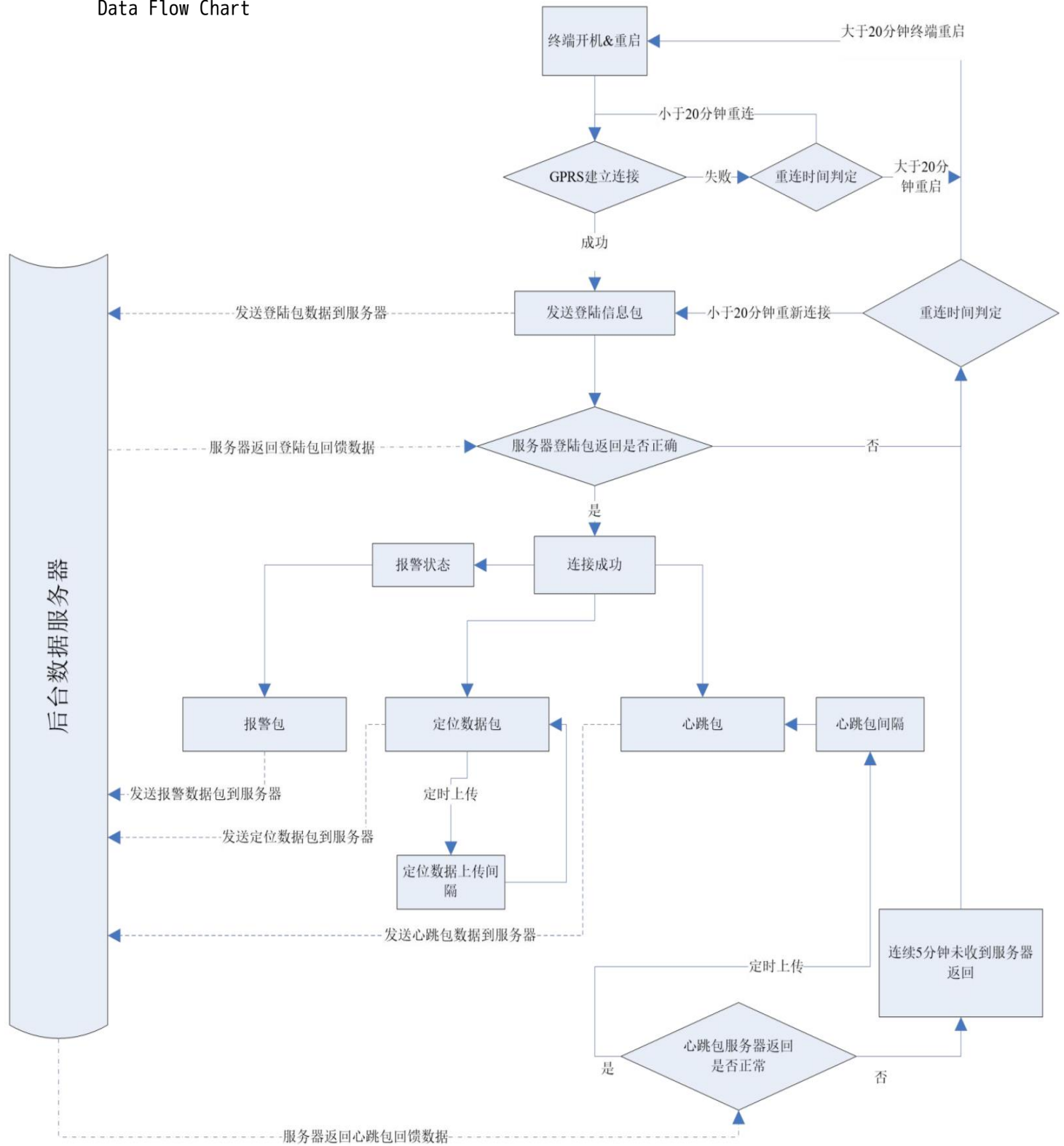
4. The connection is judged to be abnormal, repeat 3 times to establish the connection after sending the login information packet or the status information packet can not receive the data packet of the server response, the terminal starts the timed restart function, and the timing time is 10 minutes, if the terminal successfully establishes the connection with the server and receives the data packet of the server response within 10 minutes, then the timed restart function is closed, and the terminal is not restarted, or the terminal will be restarted automatically after 10 minutes. Otherwise, the terminal will restart automatically after 10 minutes;

5. After the connection is normally established, the terminal sends a combined GPS and LBS information packet to the server at regular intervals after the GPS information is changed, and the server can set the default sending protocol by command;

6. To ensure the validity of the connection, heartbeat packets are sent to the server at fixed intervals, and the server returns a response packet to confirm;

7. For terminals without registered IMEI numbers, the server should reply to the login request response and heartbeat packet response, and do not disconnect directly. (If you disconnect directly or do not reply, it will cause the terminal to reconnect constantly and the GPRS traffic will be consumed seriously).

Data Flow Chart



大于20分钟终端重启 终端开机&重启 —小于20分钟重连— 大于20分钟重启 GPRS建立连接 重连时间判定失败 成功 发送登陆信息包 重连时间判定 小于20分钟重新连接 -发送登陆包数据到服务器- 服务器登陆包返回是否正确 -服务器返回登陆包回馈数据- 是 连接成功 报警状态 后台数据服务器 心跳包间隔 定位数据包 报警包 心跳包 定时上传 发送报警数据包到服务器-- 发送定位数据包到服务器 定位数据上传间隔 ---发送心跳包数据到服务器--- 连续5分钟未收到服务器 一定时上传返回 心跳包服务器返回是否正常 是 否 ---服务器返回心跳包回馈数据-	Greater than 20 minutes of terminal restart Terminal boot & restart -Less than 20 min reconnection- Greater than 20 min to restart GPRS connection establishment Reconnection time determination failed success Send a login packet Re-connection time determination Less than 20 min to reconnect -Send the landing packet data to the server- Is the server landing package returned correctly -The server returns the landing package feedback data- Yes Connection success Alarm status Background data server Heartbeat interval Locate the packet The alarm package Heartbeat bag Timed upload Send an alarm data packet to the server- - Send the location packet to the server Location of the data upload interval ---Send the heartbeat packet data to the server--- Server not received for 5 minutes Always upload back Whether the heartbeat packet server returns is normal Yes NO ---The server returns the heartbeat packet feedback
---	--

4.1 Starting position

Fixed value, unified hex 0x78 0x78.

Note: A small number of protocol numbers, beginning with 0x79 0x79

4.2. package length

Length = Protocol Number + Message Content + Message Sequence Number + Error Check, totaling (5+N) Byte, since the message content is an indeterminate length field.

4.3. agreement number

typology	(be) worth
login pack	0x01
Positioning data (GPS/LBS merged)	0x12
Status messages (heartbeat packets)	0x13
String messages (in response to server-side commands)	0x15
warning message	0x16
Query address information (GPS)	0x1A
ICCID number to report platform information	0x94
Recorded documents reporting platform information	0x8D
issue instructions	0x80

4.4 Information content

According to the different applications, corresponding to the corresponding "protocol number", to determine the specific content.

4.5. information sequence numbers

The sequence number of the first GPRS data (including status packets and GPS, LBS, etc. packets) sent after power on is '1', and the sequence number of every data (including status packets and GPS, LBS packets) sent after that is automatically increased by 1.

4.6 Error checking

The terminal or server can use the check digit to determine whether the received information is wrong or not. In order to prevent data errors during transmission, an error check is added to prevent data misuse, which increases the safety and efficiency of the system. The error check code adopts the CRC-ITU calibration method.

The CRC-ITU value of the data in the part of the protocol body from the "packet length" to the "message sequence number" (including the "packet length", "message sequence number", "packet length", "message sequence number" and "message sequence number").) in the protocol body from "Packet Length" to "Message Sequence Number" (including "Packet Length" and "Message Sequence Number").

If the receiver receives a message with a CRC error in the calculation, it ignores it and discards the packet.

4.7 Stop bits

Fixed value, unified hex 0x0D 0x0A.

5. Terminal to the server to send packets details

Separate explanations of common packets sent and returned by the server

5.1. Logon Information Package (0x01)

5.1.1. terminal sends packet to server

The login information packet is used to confirm to the server that the connection is established properly and submit the terminal ID to the server.

specification		Length (Byte)	typical example
Login Information Package (18 Byte)	starting position	2	0x78 0x78
	Packet length	1	0x0D
	protocol number	1	0x01
	Terminal ID	8	0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45
	Message Serial Number	2	0x00 0x01
	error checking	2	0x8C 0xDD
	stop bit	2	0x0D 0x0A

5.1.1.1 Starting positions

See packet format 4.1 for details

5.1.1.2. package lengths

See packet format 4.2 for details

5.1.1.3. protocol no.

See packet format 4.3 for details

5.1.1.4 Terminal IDs

The terminal ID uses a 15-digit IMEI number.

e.g. 123456789012345.

The terminal ID is: 0x01 0x23 0x45 0x67 0x89 0x01 0x23 0x45

5.1.1.5 Information sequence numbers

See packet format 4.5 for details

5.1.1.6. error checking

See packet format 4.6 for details

5.1.1.7 Stop bits

See packet format 4.7 for details

5.1.2. server response packets

specification		Length (Byte)	typical example
response	starting	2	0x78 0x78

kit (10 Byte)	position		
	Packet length	1	0x05
	protocol number	1	0x01
	Message Serial Number	2	0x00 0x01
	error checking	2	0XD9 0xDC
	stop bit	2	0x0D 0x0A

5.1.2.1 Starting positions

See packet format 4.1 for details

5.1.2.2. package lengths

See packet format 4.2 for details

5.1.2.3. agreement number

The protocol number in the response packet is the same as the protocol number of the packet sent by the terminal

5.1.2.4. information sequence numbers

See packet format 4.5 for details

5.1.2.5. error checking

See packet format 4.6 for details

5.1.2.6 Stop bits

See packet format 4.7 for details

5.2. Positioning packets (GPS, LBS combined packet: 0x12)

5.2.1. terminal sending location packets to server

specification			Length (Byte)	typical example
Information content	starting position		2	0x78 0x78
	Packet length		1	0x1F
	protocol number		1	0x12
	Date Time		6	0x13 0x08 0x1D 0x11 0x0C 0x10
	GPS information	Number of GPS information satellites	1	0xCB
		longitude	4	0x02 0x7A 0xCF 0xEB
		longitudes	4	0xCC 0x46 0x58 0x49
		tempo	1	0x10
		Heading, status	2	0x15 0x4C
	LBS information	MCC	2	0x01 0xCC
		MNC	1	0x00

		LAC	2	0x28 0x7D
		Cell ID	3	0x00 0x1F 0xB8
	product key (software)		2	0x00 0x03
	error checking		2	0x9D 0xDC
	end position		2	0x0D 0x0A

5.2.1.1 Starting positions

See packet format 4.1 for details

5.2.1.2. package lengths

See packet format 4.2 for details

5.2.1.3. agreement number

See packet format 4.3 for details

5.2.1.4 Date and time

specification	Length	typical
surname	1	0x16
moon	1	0x01
date	1	0x1D
hour	1	0x11
ingredient	1	0x0C
unit of angle	1	0x10

e.g. January 29, 2022, 17:12:16

Calculation: 22 (decimal) = 16 (hexadecimal)

01 (decimal) = 01 (hexadecimal)

29 (decimal) = 1D (hexadecimal)

17 (decimal) = 11 (hexadecimal)

12 (decimal) = 0C (hexadecimal)

16 (decimal) = 10 (hexadecimal)

Then the value is: 0x16 0x01 0x1D 0x11 0x0C 0x10

GPS position packages with date time in 0 time zone

5.2.1.5. Number of satellites involved in positioning

1Byte converted to binary is 8Bit, the first 4Bit is the length of the GPS information, the last 4Bit is the current number of satellites of the device, the maximum value of the number of satellites is 15, if more than 15, then 15 will be reported.

Note: The length includes the 1Byte occupied by itself.

Example: A value of 0xCB means that the length of the GPS message is 12 and the number of satellites involved in the positioning is 11.

5.2.1.6. latitude

Occupies 4 bytes and represents the latitude value of the location data. The value ranges from 0 to 162000000, indicating the range of 0 degrees to 90 degrees, and the conversion method is as follows.

Convert the latitude and longitude values output from the GPS module into decimal

numbers in cents; then multiply the converted decimal numbers by 30,000 and convert the multiplication result into hexadecimal numbers.

E.g. $22^{\circ} 32.7658' = (22 \times 60 + 32.7658) \times 30000 = 40582974$, then convert to hexadecimal number

40582974 (decimal) = 26B3F3E (hex)

The last value is 0x02 0x6B 0x3F 0x3E.

Note: The uploaded latitude and longitude are all positive numbers, if they are negative take the absolute value

5.2.1.7 Longitude

Occupies 4 bytes and represents the longitude value of the positioning data. The value ranges from 0 to 324000000, indicating a range of 0 degrees to 180 degrees.

The conversion method is the same as that for latitude.

5.2.1.8.

Occupies 1 byte, indicates the operating speed of GPS, the value ranges from 0x00 to 0xFF, which means the range is 0 to 255 kilometers per hour.

As:

0x00 represents 0 km/h;

0x10 represents 16 km/h;

0xFF stands for 255 kilometers per hour.

5.2.1.9 Status heading

Occupies 2 bytes, converted to binary, the first six bits of the first byte indicate the device status, the last two bits and the eight bits of the second byte indicate the running direction of the GPS, indicating the range of 0~360, the unit: degrees, with due north as 0 degrees, clockwise.

The first six bits of the 8-bit binary of the first byte indicate the status, and the next two bits and the 8 bits of the second byte, totaling 10 bits of binary, indicate the heading degree.

BYTE_1	Bit7	No definition
	Bit6	(not in use or ACC ON/OFF)
	Bit5	GPS real-time/differential positioning
	Bit4	GPS positioning or not
	Bit3	Longitude East, Longitude West
	Bit2	South, North
	Bit1	directiondirection (a ship or plane is heading in)
	Bit0	
BYTE_2	Bit7	
	Bit6	
	Bit5	
	Bit4	

	Bit3	
	Bit2	
	Bit1	
	Bit0	

Note: The status information in the packet are all the status at the moment recorded by the time bit in the packet.

For example, the value 0x15 0x4C becomes binary 00010101 01001100.

BYTE_1 Bit7	0 (not defined yet)	
BYTE_1 Bit6	0 (ACC OFF) --0: ACC OFF, 1: ACC ON (some items do not use this bit)	
BYTE_1 Bit5	0 (real-time GPS) - 0: real-time GPS, 1: differential positioning	
BYTE_1 Bit4	1 (GPS positioned) - 0: GPS not positioned, 1: GPS positioned	
BYTE_1 Bit3	0 (east longitude) - 0: east longitude, 1: west longitude	
BYTE_1 Bit2	1 (north latitude) - 0: south latitude, 1: north latitude	
BYTE_1 Bit1	0	
BYTE_1 Bit0	1	
BYTE_2 Bit7	0	
BYTE_2 Bit6	1	
BYTE_2 Bit5	0	
BYTE_2 Bit4	0	
BYTE_2 Bit3	1	
BYTE_2 Bit2	1	
BYTE_2 Bit1	0	
BYTE_2 Bit0	0	

Heading 332° (0101001100 binary to decimal 332)

This means that the GPS has been located and the real-time GPS, latitude, longitude, and heading is 332°.

5.2.1.10 MCC

Mobile Country Code (MCC).

For example: China's mobile country number is 460 (decimal): 0x01 0xCC (decimal 460 to hexadecimal, hexadecimal less than four left complement 0).

The range of values here is 0x0000 to 0x03E7.

5.2.1.11 MNC

Mobile Network Code(MNC)

For example, 0x00 for China Mobile.

5.2.1.12 LAC

The Location Area Code (LAC) is contained in the LAI and consists of two bytes encoded in hexadecimal. The available range is 0x0001-0xFFFFE, code groups 0x0000 and 0xFFFF are not available (see GSM specifications 03.03, 04.08 and 11.11). A location area can contain one or more cells.

5.2.1.13. Cell ID

Mobile Base Station Cell Tower ID (Cell ID), value range 0x000000 to 0xFFFFFFFF

5.2.1.14. information sequence numbers

See packet format 4.5 for details

5.2.1.15. error checking

See packet format 4.6 for details

5.2.1.16. Stop bits

See packet format 4.7 for details

5.2.2. server response

The server does not need to respond to this message

Note: 0x12 GPS/LBS information package will be saved when GSM signal is abnormal and then uploaded to the platform later; the uploading method is not real-time uploading, but uploading the retained positioning points after the device is stationary. The time in the positioning packet should be used for track playback, and the time in the positioning packet should not be displayed when the real-time display is less than 5 minutes from the current time.

5.3. Heartbeat kit

Heartbeat packets are packets that maintain the connection between the terminal and the server

5.3.1. terminal sends heartbeat packet to server (0x13)

specification		Length (Byte)	typical example
Information content	starting position	2	0x78 0x78
	Packet length	1	0x0A
	protocol number	1	0x13
	status message	Terminal Information	0x4E
		voltage level	0x06
		GSM signal strength	0x64
		External Voltage	0x0C
		multilingualism	0x02
	product key (software)		0x00 0x03
	error checking		0xFF 0x4F
	end position		0x0D 0x0A

5.3.1.1 Starting positions

See packet format 4.1 for details

5.3.1.2. package lengths

See packet format 4.2 for details

5.3.1.3. protocol no.

See packet format 4.3 for details

5.3.1.4 Terminal information

Occupies 1 byte, converted to binary, used to represent the status information of the terminal, 1 byte is regarded as 8 bits, the lowest bit is 0 bit, the highest bit is 7 bit, the high bit is transmitted first, and then the low bit is transmitted. The specific meanings of each representative are as follows:

classifier for honorific people		Code Meaning
BYTE	Bit7	1: Oil-electric disconnection
		0: Oil and electricity connected
	Bit6	1: GPS has been located
		0: GPS not positioned
	Bit5-Bit3	100: SOS alarm
		011: Low battery alarm
		010: Power failure alarm
		001: Vibration alarm
		000: Normal

	Bit2	1: Connected to external power supply
		0: No external power supply connected
	Bit1	1: ACC ON
		0: ACC OFF
	Bit0	1: Setting up defenses
		0: Disarming

Example: 0x4E, the corresponding binary is 01001110

Indicates that it is in the disarmed state, ACC ON, connected to external power, vibration alarm, GPS has been located, and oil and electricity are connected.

5.3.1.5 Voltage classes

There are 7 voltage levels in total, ranging from 0 to 6, indicating the voltage magnitude from low to high.

0: Low power shutdown;

1: Not enough power to make phone calls, text messages, etc;

2: Low power is too low;

3-6: All can be used normally, just depending on the amount of power.

5.3.1.6. GSM signal strength

Range of GSM information: 0-100; converted to decimal, the larger the value, the stronger the GSM signal

0: Indicates no signal

100: full signal

5.3.1.7. external voltage + language

Example: the external power supply voltage is 30 volts (30V), the current language bit of the terminal. :

Chinese:0x1E 0x01;

English:0x1E 0x02;

Note: The reserved expansion bits of the S11/S11C/W15L are not used as language bits, but are used to identify the operating mode of the device and the device sleep state.

Work mode:

0x00 = Intelligent hibernation mode;

0x01 = normal mode;

0x02 = deep hibernation mode;

0x03 = Timed return mode;

0x04 = Remote switching mode;

The device is dormant:

0x00 for normal device operation;

When the device is hibernating, the device hibernation status bit is the same as the operating mode bit, for example, in deep hibernation mode, the reserved extended bit is 0x02 0x02 when the device is hibernating.

5.3.1.8 Information sequence numbers

See packet format 4.5 for details

5.3.1.9 Error checking

See packet format 4.6 for details

5.3.1.10. Stop bits

See packet format 4.7 for details

5.3.2. Server Response to Heartbeat Packets

specification		Length (Byte)	typical example
Information content	starting position	2	0x78 0x78
	Packet length	1	0x05
	protocol number	1	0x13
	information sequence	2	0x00 0x01
	error checking	2	0xE9 0xF1
	stop bit	2	0x0D 0x0A

The server responds to a packet with an empty message content after receiving the packet from the terminal.

Note: The "message sequence number" in the packet must be the same as the "message sequence number" sent by the responding terminal.

5.3.2.1 Starting positions

See packet format 4.1 for details

5.3.2.2 Package length

See packet format 4.2 for details

5.3.2.3 Protocol number

See packet format 4.3 for details

5.3.2.4 Information sequence numbers

See packet format 4.5 for details

5.3.2.5 Error checking

See packet format 4.6 for details

5.3.2.6 Stop bits

See packet format 4.7 for details

5.4. Alarm Package (0x16)

5.4.1. terminal sends alarm packet to server

Specification		Length (Byte)	Typical example
Information content	starting position	2	0x78 0x78
	packet length	1	0x25
	protocol number	1	0x16
	Date Time	6	0x13 0x08 0x1D 0x11 0x0C 0x10
	GPS information	number of GPS information satellites	0xCB
		longitude	0x02 0x7A 0xCF 0xEB
		longitudes	0xCC 0x46 0x58 0x49
		tempo	0x10
		heading, status	0x15 0x4C
	LBS information	LBS length	0x09
		MCC	0x01 0xCC
		MNC	0x00
		LAC	0x28 0x7D
		Cell ID	0x00 0x1F 0xB8
	status message	terminal Information	0x4E
		voltage level	0x04
		GSM signal strength	0x64
		give a warning	0x00
		multilingualism	0x02
	product key (software)		0x00 0x03
	error checking		0x8C 0x59
	end position		0x0D 0x0A

5.4.1.1 Starting positions

See packet format 4.1 for details

5.4.1.2 Package length

See packet format 4.2 for details

5.4.1.3 Protocol number

See packet format 4.3 for details

5.4.1.4 Date time

See packet format 5.2.1.4 for details.

5.4.1.5 Number of GPS satellites

See Packet Format 5.2.1.5 for details.

5.4.1.6 Latitude

See packet format 5.2.1.6 for details.

5.4.1.7 Longitude

See packet format 5.2.1.7 for details.

5.4.1.8 Speed

See packet format 5.2.1.8 for details.

5.4.1.9 Heading, status

See packet format 5.2.1.9 for details.

5.4.1.10 LBS lengths

The length of the LBS message content, the length is fixed: 0x09

5.4.1.11 MCC

See packet format 5.2.1.10 for details.

5.4.1.12 MNC

See packet format 5.2.1.11 for details.

5.4.1.13 LAC

See packet format 5.2.1.12 for details.

5.4.1.14 Cell ID

See packet format 5.2.1.13 for details.

5.4.1.15 Terminal information

See packet format 5.3.1.4 for details.

5.4.1.16 Battery voltage rating

See packet format 5.3.1.5 for details.

5.4.1.17 GSM signal strength

See packet format 5.3.1.6 for details.

5.4.1.18 Alarm types

Alarm type	0x00: Normal
	0x01: SOS alarm
	0x02: Power failure alarm
	0x03: Vibration alarm
	0x04: Into the fence alarm
	0x05: Out of fence alarm
	0x06: Speed alarm
	0x09: Displacement alarm
	0x0E: Low battery alarm
	0xFE: ACC flameout alarm
	0xFF: ACC Ignition Alarm

5.4.1.19 Language

Chinese: 0x01

English: 0x02

5.4.1.20 Information sequence numbers

See packet format 4.5 for details

5.4.1.21 Error checking

See packet format 4.6 for details

5.4.1.22 Stop bits

See packet format 4.7 for details

5.4.2. server sends alarm packet reply to terminal

specification		Length (Byte)	typical example
Information	starting position	2	0x78 0x78

content	Packet length	1	0x05
	protocol number	1	0x16
	information sequence	2	0x00 0x05
	error checking	2	0x96 0x68
	stop bit	2	0x0D 0x0A

5.4.2.1 Starting positions

See packet format 4.1 for details

5.4.2.2 Package length

See packet format 4.2 for details

5.4.2.3 Protocol number

See packet format 4.3 for details

5.4.2.4 Information sequence numbers

See packet format 4.5 for details

5.4.2.5 Error checking

See packet format 4.6 for details

5.4.2.6 Stop bits

See packet format 4.7 for details

5.5. LBS multi-base station positioning information (0x18)

specification			Length (Byte)
Information content	Date Time		6
	LBS information	MCC	2
		MNC	1
		LAC	2
		MCI	2
		MCISS	1
		MCI1	2
		MCISS1	1
		MCI2	2
		MCISS2	1

		MCI3	2
		MCISS3	1
		MCI4	2
		MCISS4	1
		MCI5	2
		MCISS5	1
		MCI6	2
		MCISS6	1
	Reserve Expansion Position		N

5.5.1 Date and time

Same as the previous protocol description.

5.5.2 MCC

Same MCC description as in the previous LBS message

5.5.3 MNC

Same MNC description as in the previous LBS message

5.5.4 LAC

Same LAC description as in the previous LBS message

5.5.5 MCI (Main Cell ID)

Mobile Base Station Cell Tower ID (Cell ID), value range 0x0000 to 0xFFFF.

5.5.6. MCISS (Main Cell ID Signal Strength)

Main cell signal strength, value range is 0x00~0xFF, 0x00 signal is the weakest, 0xFF signal is the strongest.

5.5.7 MCI1-6 (Near Cell ID)

Neighboring cell base station codes, 6 in total, with values ranging from 0x0000 to 0xFFFF.

5.5.8. NCISS1-6 (Near Cell ID Signal Strength)

Neighboring cell BTS signal strength, corresponding to each of the 6 neighboring cell BTS codes. The value range is 0x00 to 0xFF.

The absolute value of the signal strength is used here, and the value should be taken with a negative sign attached.

5.5.9 Reservation of expansion spaces

This item is currently empty

5.6. Query address (0x1A)

5.6.1 Terminal sending query address packet to server

specification	Length (Byte)	typical example
---------------	---------------	-----------------

	Data content	Type of information	1	00 : External voltage 01~03 : (Customized) 04: Terminal state synchronization 05: Door status 08: Self-test parameters 09: Positioning satellite information 0A: ICCID information to be added
		Data content	N	The content varies depending on the type of information, 0A detailed in the table below
		ICCID	10	Example: If the ICCID number is 12345123456789123456, the terminal ID is: 0x12 0x34 0x51 0x23 0x45 0x97 0x89 0x12 0x34 0x56
	Message Serial Number		2	Auto +1 for every data sent from power up
	error checking		2	CRC-ITU value from "packet length" to "message sequence number".
	end position		2	0x0D 0x0A

When the type is 0A, in addition to transmitting ICCID information, the transmission is hexadecimal

IMEI	8	Example: If the IMEI number is 123456789123456, the terminal ID is: 0x01 0x23 0x45 0x67 0x89 0x12 0x34 0x56
IMSI	8	Example: IMSI number 123456789123456, then the terminal ID is: 0x01 0x23 0x45 0x67 0x89 0x12 0x34 0x56
ICCID	10	Example: If the ICCID number is 12345123456789123456, then the terminal ID is: 0x12 0x34 0x51 0x23 0x45 0x97 0x89 0x12 0x34 0x56

Example of a device uploading an ICCID number:

79790020940a03580910880015580460041990205313898607b91117301203130009a0720d0a

7979 0020 94 0a 0358091088001558 0460041990205313

Start Bit Length Protocol No. Type IMEI No. IMSI No.

898607b9111730120313 0009 a072 0d0a

ICCID number Message sequence Error check Stop bit

Note: This agreement, the server does not need to reply

5.10. Recording protocol package (0x8D)

Send the recording packet to the server after the device recording, applicable to S709,S11,S11C,W15L devices

5.10.1. the terminal sends a packet to the server:

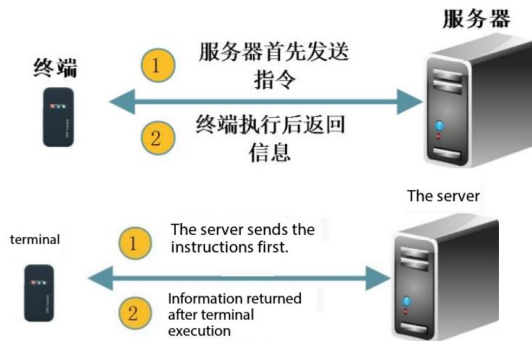
specification	Length (Byte)	descriptive
---------------	---------------	-------------

Information content	starting position		2	Fixed value: 0x79 0x79
	Packet length		2	Length = protocol number + message content + message sequence number + error checking
	protocol number		1	Fixed value: 0x8D
	Content of recordings	Document type	1	Fixed value: 0x00
		Total length of document	4	Total length of transmitted files
		Total number of packages	1	The total number of packages into which the file is transferred
		The current package sequence of the file	1	The current package sequence of the transfer file
		Current content length	2	Length of data from the beginning of the transmission segmentation
		Data content	M	Segmented packet content
	Message Serial Number		2	Auto +1 for every data sent from power up
	error checking		2	CRC-ITU value from "packet length" to "message sequence number".
	end position		2	0x0D 0x0A

5.10.2. the server responds to the terminal:

specification		Length (Byte)	descriptive
Information content	starting position	2	Fixed value: 0x79 0x79
	Packet length	2	Length = protocol number + message content + message sequence number + error checking
	protocol number	1	Fixed value: 0x8D
	Receive Status Flag	1	0x00:receive normal 0x01:receive error
	Message Serial Number	2	Auto +1 for every data sent from power up
	error checking	2	CRC-ITU value from "packet length" to "message sequence number".
	end position	2	0x0D 0x0A

Every time the device sends a recording file packet, it needs to receive a response from the server to receive it before it continues to send the next packet, if it sends the same packet for three times in a row and does not receive a response, it will give up this recording file sending.



6. Server sends packets to terminals

6. 1. Server send command (0x80)

specification		Length (Byte)
starting position		2
Packet length		1
protocol number		1
Information content	instruction length	1
	Server Flags	4
	Instruction content	N
Message Serial Number		2
error checking		2
stop bit		2

6.1.1 Starting positions

See packet format 4.1 for details

6.1.2 Package length

See packet format 4.2 for details

6.1.3. protocol number

Fixed value: 0x80

6.1.4. Instruction length

Instruction length = server flag bit + instruction content length

For example, in terms of byte length, 0x0A, which means that the content of the instruction occupies 10 bytes

6.1.5. server flag bits

Reserved for server identification, the terminal returns the received data binary as is in the return packet

6.1.6. command contents

ASCII representation as a string, command content compatible with SMS commands

6.1.7 Information sequence numbers

See packet format 4.5 for details

6.1.8 Error checking

See packet format 4.6 for details

6.1.9 Stop bits

See packet format 4.7 for details

6.2. terminal return packet (0x15)

specification		Length (Byte)
starting position		2
Packet length		1
protocol number		1
Information content	instruction length	1
	Server Flags	4
	Instruction content	N
	multilingualism	2
Message Serial Number		2
error checking		2
stop bit		2

6.2.1 Starting position

See packet format 4.1 for details

6.2.2. package length

See packet format 4.2 for details

6.2.3. protocol number

The terminal responds to the command sent by the server, and the packet format is the same as that of "Command sent by the server to the terminal", but the protocol number is different, and 0x15 is used.

6.2.4. Instruction length

Instruction length = server flag bit + instruction content length

For example, in terms of byte length, 0x0A, which means that the content of the instruction occupies 10 bytes

6.2.5. server flag bits

The terminal returns the received data binary as is in the return packet

6.2.6. command contents

ASCII representation of a string

6.2.7. languages

Language of the terminal

Chinese:0x00 0x01

English:0x00 0x02

6.2.8 Information sequence numbers

See packet format 4.5 for details

6.2.9 Error checking

See packet format 4.6 for details

6.2.10 . Stop Bits

See packet format 4.7 for details

6.2.11 Example of Server Command Issuance

6.2.1.1.1 Disconnection of oil and electricity

```

UDS 11 Disconnection of oil and electricity
UDS command Format:
UDS 0000000#
Function description: Cut off the vehicle fuel and electric control circuits
Return information:
Successful return: DYS=Success!
Failure returns: DYS=Invalid Fix!
or DYS=Speed Limit, Speed 40km/h
Example platform down: 78 78 15 80 0F 00 01 A9 58 44 59 44 2C 30 30 30 30 30 30 30
Boded: DYS 0000000#
00 18 77 00 0A
Device response: 78 78 18 15 10 00 01 A9 58 44 59 44 3D 53 75 63 63 65 73 73 21 00 02
Boded: DYS=Success!

```

6.2.11.2 Restoration of oil and electricity

SMS Command Format:
 HFYD_000000#
 Function Description: Turns on the vehicle's fuel and electric control circuits
 Return information:
 Successful return: HFYD=Success!
 Failure to return: HFYD=Fail!
 Platform issuance and equipment response is similar to DYD

6.2.11.3 Viewing locations

Command Format: **DWXX 000000#**

Functional Description:
Get Location Information command. Both the cell phone user and the SMS server can use this command to get location information.

Return Information:
Successful return: DWXX=Lat:<South/North Latitude> Lon:<East/West Longitude> Course:<Angle> Speed:<Speed>, Date Time:<Time>
Failure to return: DWXX=Command Error!

Example:
Lat:N23d5.1708m Lon:E114d23.6212m Course:120 Speed:53.02 Date Time:08-09-12 14:52:36
Meaning: Latitude:23 degrees 5.1708 minutes North; Longitude 114 degrees 23.6212 minutes East; Angle: 120 degrees; Speed: 53.02 kilometers per hour; Time and Date: 12/9/08 14:52:36 seconds

Note: If the terminal is not positioned successfully, it returns:
Lat, Lon, Course, Speed, Date Time:-

Platform Issuance and equipment response format similar to DYD

```

Upload: 78 78 0D 01 08 68 02 00 41 12 74 64 00 1E A0 1E 0D 0A
Next: 78 78 05 01 00 2B 15 06 84 0D 0A
Upload: 78 78 1F 00 12 15 06 11 07 38 35 CF 02 6C 6C F7 0C 2B 15 20 0B 14 76 01 CC 00 26
33 00 0E 81 00 1D B3 A8 0D 0A
Upload: 78 78 0A 13 44 06 03 0B 02 00 1F B4 E2 0D 0A
Next: 78 78 05 13 00 2B 67 A9 0D 0A
Upload: 78 78 09 00 20 94 0A 68 02 00 41 12 74 64 04 60 04 04 21 92 04 69 89 86 02 B4 03
19 C0 01 04 69 0D 20 94 0A C9 0D 0A
Upload: 78 78 25 16 16 06 11 08 12 25 C0 02 6C 69 7B 0C 37 1C 92 00 04 00 08 01 CC 00 26
33 00 10 04 70 04 04 02 02 00 11 41 56 0D 0A
Upload: 78 78 04 12 16 06 11 08 10 10 34 01 CC 00 26 33 0E 8B 35 0F 5D 2B 10 04 2A 0E 89
2A 1A 19 29 0E EC 25 16 06 26 00 0D 6E 60 0D 0A
下发: 78 78 2E 1A 16 06 11 08 15 1A CF 02 6C 6E E6 0C 37 14 0C 00 14 00 20 20 20 20
20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
Upload: 78 78 15 80 0F 00 01 A9 58 44 59 44 2C 30 30 30 30 30 30 30 23 00 A0 DC F1
0D 0A
Upload: 78 78 18 15 10 00 01 A9 58 44 59 44 3D 53 75 63 63 65 73 73 21 00 02 00 0A A2 E4
0D 0A

```

7. Attachment A A U CRC-ITU lookup table algorithm C C language code snippet

CRC-ITU lookup table algorithm C code snippet

```
static const U16 crctab16[] =
```

```
{
0x0000, 0x1189, 0x2312, 0x329b, 0x4624, 0x57ad, 0x6536, 0x74bf.
0x8c48, 0x9dc1, 0xaf5a, 0xbed3, 0xca6c, 0xdbe5, 0xe97e, 0xf8f7.
0x1081, 0x0108, 0x3393, 0x221A, 0x56a5, 0x472c, 0x75b7, 0x643e.
0x9cc9, 0x8d40, 0xbfdb, 0xae52, 0xdaed, 0xcb64, 0xf9ff, 0xe876.
0x2102, 0x308b, 0x0210, 0x1399, 0x6726, 0x76af, 0x4434, 0x55bd.
0xad4a, 0xbcc3, 0x8e58, 0x9fd1, 0xeb6e, 0xfae7, 0xc87c, 0xd9f5.
0x3183, 0x200a, 0x1291, 0x0318, 0x77a7, 0x662e, 0x54b5, 0x453c.
0xbdcb, 0xac42, 0x9ed9, 0x8f50, 0xfbef, 0xea66, 0xd8fd, 0xc974.
0x4204, 0x538d, 0x6116, 0x709f, 0x0420, 0x15a9, 0x2732, 0x36bb.
0xce4c, 0xdfc5, 0xed5e, 0xfcd7, 0x8868, 0x99e1, 0xab7a, 0xbaf3.
0x5285, 0x430c, 0x7197, 0x601E, 0x14a1, 0x0528, 0x37b3, 0x263a.
0xdecd, 0xcf44, 0xfddf, 0xec56, 0x98e9, 0x8960, 0xbbfb, 0xaa72.
0x6306, 0x728f, 0x4014, 0x519d, 0x2522, 0x34ab, 0x0630, 0x17b9.
0xef4e, 0xfec7, 0xcc5c, 0xddd5, 0xa96a, 0xb8e3, 0x8a78, 0x9bf1.
0x7387, 0x620e, 0x5095, 0x411c, 0x35a3, 0x242a, 0x16b1, 0x0738.
0xffcf, 0xee46, 0xdcdd, 0xcd54, 0xb9eb, 0xa862, 0x9af9, 0x8b70.
0x8408, 0x9581, 0xa71a, 0xb693, 0xc22c, 0xd3a5, 0xe13e, 0xf0b7.
0x0840, 0x19c9, 0x2B52, 0x3adb, 0x4e64, 0x5fed, 0x6d76, 0x7cff.
0x9489, 0x8500, 0xb79b, 0xa612, 0xd2ad, 0xc324, 0xf1bf, 0xe036.
0x18c1, 0x0948, 0x3bd3, 0x2a5a, 0x5ee5, 0x4f6c, 0x7df7, 0x6c7e.
0xa50a, 0xb483, 0x8618, 0x9791, 0xe32e, 0xf2a7, 0xc03c, 0xd1b5.
0x2942, 0x38cb, 0x0a50, 0x1bd9, 0x6f66, 0x7eef, 0x4c74, 0x5dfd.
0xb58b, 0xa402, 0x9699, 0x8710, 0xf3af, 0xe226, 0xd0bd, 0xc134.
0x39c3, 0x284a, 0x1ad1, 0x0b58, 0x7fe7, 0x6e6e, 0x5cf5, 0x4d7c.
0xc60c, 0xd785, 0xe51e, 0xf497, 0x8028, 0x91a1, 0xa33a, 0xb2b3.
0x4a44, 0x5bcd, 0x6956, 0x78df, 0x0c60, 0x1de9, 0x2f72, 0x3efb.
0xd68d, 0xc704, 0xf59f, 0xe416, 0x90a9, 0x8120, 0xb3bb, 0xa232.
0x5ac5, 0x4b4c, 0x79d7, 0x685e, 0x1ce1, 0x0d68, 0x3ff3, 0x2e7a.
0xe70e, 0xf687, 0xc41c, 0xd595, 0xa12a, 0xb0a3, 0x8238, 0x93b1.
0x6b46, 0x7acf, 0x4854, 0x59dd, 0x2d62, 0x3ceb, 0x0e70, 0x1ff9.
0xf78f, 0xe606, 0xd49d, 0xc514, 0xb1ab, 0xa022, 0x92b9, 0x8330.
0x7bc7, 0x6a4e, 0x58d5, 0x495c, 0x3de3, 0x2C6a, 0x1ef1, 0x0f78.
};
// Calculate the 16-bit CRC for the given length of data.
U16 GetCrc16(const U8* pData, int nLength)
{
```

```
U16 fcs = 0xffff; // Initialization
while(nLength>0){
fcs = (fcs >> 8) ^ crctab16[(fcs ^ *pData) & 0xff];
nLength --;
pData++;
}
return ~fcs; // invert
}
```